IT3081 STATISTICAL MODELLING — MASTER VIVA DEFENSE GUIDE

Client: BladeGen Tech (F&B; Chain POS Analytics) | **Course:** IT3081 Statistical Modelling
Structure: Slide-by-Slide 3-Tier Defense Question Bank (Foundational, Intermediate, Adversarial)

Tier 1: Foundational (Easy)	Verifies exact numbers, basic statistical definitions, formulas, and syllabus requirements.
Tier 2: Intermediate (Deep)	Probes mathematical assumptions, model comparison rationale, and trade-offs.
Tier 3: Adversarial (Trap)	Tackles edge cases, methodology defense, and commercial operational constraints.

SLIDE 01 FOCUS

Slide 01: Title & Strategic Consultancy Context — *Analytical Unit, Project Scope & Commercial Setting*

[TIER 1: FOUNDATIONAL] What is the primary analytical unit of your dataset and why?

DEFENSE: The primary analytical unit is the individual purchased item line (29,465 records). While multi-item baskets share an OrderNo (20,476 unique orders), checkout upselling occurs item-by-item as patrons order.

Audit Evidence: Slide 1 & Slide 4; 01_source_audit.csv.

[TIER 2: INTERMEDIATE] Why is this project framed as statistical consultancy rather than a purely academic exercise?

DEFENSE: The analytics directly address two live commercial needs for BladeGen Tech: (1) real-time category upselling at cash registers, and (2) supply chain demand forecasting for perishable bakery and dairy logistics.

Audit Evidence: Executive Summary; Slide 2.

[TIER 3: ADVERSARIAL / EDGE-CASE] If items within an order are correlated, does item-level modeling violate independence?

DEFENSE: Item lines within the same order are conditioned on common basket context. We engineered order-level features (BasketItems and BasketSpend) to capture basket scale, and we identify grouped order-level cross-validation as an essential operational next step.

Audit Evidence: Slide 15; Model Validation Limitations.

SLIDE 02 FOCUS

Slide 02: Industry Problem Formulation & Business Decisions — *Business Questions & Spend Regression Clarification*

[TIER 1: FOUNDATIONAL] State clearly the two distinct business questions formulated for BladeGen Tech.

DEFENSE: Question 1 (Register Upselling): 'Which product category—Beverage, Food, Deals, or Merchandising—is an ordering customer most likely to purchase next given observable contextual features?' Question 2 (Perishable Demand Planning): 'How many total customer orders will our 80 retail outlets receive over the next 30 days to optimize daily fresh bakery and dairy replenishment?'

Audit Evidence: Slide 2; Report Executive Summary.

[TIER 2: INTERMEDIATE] Why did you prioritize category classification over spend regression as the primary operational engine?

DEFENSE: In retail POS systems, Revenue is an exact accounting identity: Revenue = (Adjusted Rate * Qty) - Discount. Using price and quantity to predict spend constitutes circular data leakage for prospective orders. We evaluated spend regression (Ridge & LASSO) as an exploratory continuous spend baseline in the appendix, but focused primarily on multi-class category choice.

Audit Evidence: Report Task 5; Slide 30.

[TIER 3: ADVERSARIAL / EDGE-CASE] Are the 14.5% conversion and 4.5% waste numbers proven current capabilities?

DEFENSE: No, they are clearly defined proposed operational targets and benchmark KPIs for the client's phased pilot, designed to evaluate measurable commercial lift during field testing.

Audit Evidence: Slide 2; Slide 27.

SLIDE 03 FOCUS

Slide 03: Systematic Literature Review (15 Studies) — *Multi-Metric Evaluation, LDA Assumptions & Experimental Blocking*

[TIER 1: FOUNDATIONAL] How many research studies support your methodology, and what core areas do they cover?

DEFENSE: Our methodology is supported by 15 peer-reviewed studies across multi-metric classification evaluation (Chicco & Jurman 2020; Hand & Till 2001), discriminant analysis foundations (Fisher 1936; Hastie et al. 2009), missing data theory (Little & Rubin 2019), experimental design (Bradlow et al. 2023), and seasonal time series forecasting (Hyndman & Khandakar 2008).

Audit Evidence: Slide 3; Slides 31–33 Bibliography.

[TIER 2: INTERMEDIATE] Why is multi-metric evaluation critical when evaluating imbalanced multi-class problems?

DEFENSE: As Chicco & Jurman (2020) and Hand & Till (2001) demonstrate, overall accuracy in imbalanced datasets is heavily dominated by majority categories. Evaluating Balanced Accuracy, Macro-F1, Matthews Correlation Coefficient (MCC), and multiclass ROC-AUC ensures that minority categories (Deals 8.7%, Merchandising 5.4%) are accurately recognized.

Audit Evidence: Chicco & Jurman (2020); Hand & Till (2001); Slide 13.

[TIER 3: ADVERSARIAL / EDGE-CASE] Why not use deep learning or complex Transformer architectures for checkout prediction?

DEFENSE: Tabular POS transactional records lack the spatial invariance of images or sequential syntax of text. Complex neural nets risk overfitting tabular noise, require cloud API roundtrips (>150ms), and lack the deterministic closed-form stability of Linear Discriminant Analysis on edge POS registers.

Audit Evidence: Hastie et al. (2009); Slide 14.

SLIDE 04 FOCUS

Slides 04–05: Dataset Audit & Missing Value Preprocessing — *Data Quality, Demographic Missingness & Structural Indicators*

[TIER 1: FOUNDATIONAL] Describe the dimensions and time horizon of the dataset analyzed.

DEFENSE: The dataset contains 29,465 authentic transactional item records across 80 retail outlets in Western Province, Sri Lanka, spanning exactly 90 calendar days from January 1 to March 31, 2026. Target categories are Beverage (52.2%), Food (33.7%), Deals (8.7%), and Merchandising (5.4%).

Audit Evidence: Slide 4; df_clean.csv; 01_source_audit.csv.

[TIER 2: INTERMEDIATE] How were missing values detected and handled in continuous vs. categorical features?

DEFENSE: Core commercial fields (Rate, Qty, Discount, ServiceCharge, Outlet, OrderDate, PaymentMethod, OrderType) showed 100% completeness with 0% missingness. Missingness was confined exclusively to optional demographic fields (CustomerLoyaltyNumber and Age), reflecting anonymous walk-in patrons. Following Little & Rubin (2019), naive mean or mode imputation was avoided because single imputation artificially deflates sample variance and distorts covariance structures. Instead, an explicit structural indicator was derived: IsLoyaltyMember ('Yes' vs 'No').

Audit Evidence: Little & Rubin (2019); Slide 5; 01_data_cleaning.R.

[TIER 3: ADVERSARIAL / EDGE-CASE] Why didn't you perform complete-case analysis (listwise deletion) for rows with missing loyalty data?

DEFENSE: Over 62% of transactions lacked a loyalty number because walk-in patrons chose not to register. Complete-case deletion would discard over 60% of our dataset, severely biasing our sample toward registered club members who possess fundamentally different spending habits.

Audit Evidence: Little & Rubin (2019); Slide 5.

SLIDE 08 FOCUS **Slide 08: Statistical Inference Evidence (6 Hypothesis Tests)** — *Welch's t-test, Chi-Square, ANOVA & Kruskal-Wallis*

[TIER 1: FOUNDATIONAL] Which of your 6 formal hypothesis tests yielded statistically significant results at alpha = 0.05?

DEFENSE: Three tests were statistically significant: Test 1 (Welch's t-test: Foreign vs Local spend, t = 2.0266, p = 0.0427, +LKR 14.57 difference); Test 4 (Chi-Square: Payment Method x Customer Type, Chi^2 = 4,270.20, p < 0.0001, Cramer's V = 0.381); and Test 6 (Pearson Correlation: Discount vs Revenue, r = 0.0433, p < 0.0001).

Audit Evidence: inference_summary.csv; Slide 8.

[TIER 2: INTERMEDIATE] Why did you use Welch's t-test instead of Student's two-sample t-test for foreign vs. local spend?

DEFENSE: Local customers outnumber foreign patrons by more than 3 to 1 (22,442 vs 7,023) and group variances differ. As Welch (1947) demonstrated, Student's t-test relies on a pooled variance assumption that distorts Type I error rates when sample sizes differ. Welch's test uses Satterthwaite's degrees of freedom approximation, providing robust Type I error control under heteroscedasticity.

Audit Evidence: Welch, B.L. (1947, Biometrika, 34, p. 28); Slide 8.

[TIER 3: ADVERSARIAL / EDGE-CASE] In Test 3 (One-Way ANOVA across top 8 outlets), p = 0.3266, failing to reject H0. What does this imply commercially?

DEFENSE: It proves that average transaction spend is statistically homogeneous across top outlets. Store revenue differences are driven almost entirely by foot-traffic volume (transaction count), not by patrons spending significantly more per line at flagship stores.

Audit Evidence: Slide 8; inference_summary.csv.

SLIDE 10 FOCUS **Slide 10: Class Imbalance: Defending Natural Stratification** — *Class Imbalance, Resampling Strategy & Prior Preservation*

[TIER 1: FOUNDATIONAL] What is the class distribution of your target variable, and what train/test split was used?

DEFENSE: Target variable is MainCategory: Beverage (52.17%), Food (33.71%), Deals (8.73%), Merchandising (5.39%). We used an 80/20 stratified split (23,572 training rows, 5,893 testing rows), preserving natural class shares across both partitions.

Audit Evidence: Slide 10; summary_categorical.csv.

[TIER 2: INTERMEDIATE] Why was synthetic oversampling (SMOTE) deliberately not applied?

DEFENSE: As He & Garcia (2009) explain, artificially rebalancing training distributions distorts posterior class probabilities P(Y=k|X). In an operational POS recommendation setting, an oversampled model would suggest low-frequency Deals and Merchandising at artificially inflated rates, fatiguing cashiers. Furthermore, synthetic interpolation across 19 mixed dummy indicator variables risks creating artificial boundary artifacts.

Audit Evidence: He & Garcia (2009); Slide 10.

[TIER 3: ADVERSARIAL / EDGE-CASE] Doesn't class imbalance bias standard accuracy toward the majority class?

DEFENSE: Yes, which is precisely why we do not rely solely on accuracy. We evaluate models using Balanced Accuracy (macro-recall across classes), Macro-F1, and Matthews Correlation Coefficient, which are mathematically invariant to class prevalence.

Audit Evidence: Chicco & Jurman (2020); Slide 10; Slide 13.

SLIDE 14 FOCUS **Slide 14: Winning Model Selection (Linear Discriminant Analysis)** — *Multi-Metric Equilibrium, QDA Comparison & Edge Scoring*

[TIER 1: FOUNDATIONAL] Why was Linear Discriminant Analysis chosen as Champion over Multinomial Logistic Regression?

DEFENSE: LDA leads Multinomial Logit in Top-1 Accuracy (76.29% vs 76.18%, delta = 0.11 pp) and Macro-F1 (76.18% vs 75.87%), while achieving higher MCC (0.5996 vs 0.5968) and computing linear discriminant vectors via closed-form linear projections without iterative numerical optimization.

Audit Evidence: Slide 14; model_comprehensive_metrics.csv.

[TIER 2: INTERMEDIATE] Didn't Quadratic Discriminant Analysis (QDA) achieve a higher Balanced Accuracy than LDA?

DEFENSE: LDA leads in Balanced Accuracy at 73.79% (vs Multinomial 73.14%, QDA 72.56%, Naïve Bayes 72.77%). Furthermore, QDA requires estimating four separate covariance matrices, creating singularity risks in minority categories like Merchandising (n=1,588). LDA pools covariance across all classes, providing stable, regularized decision boundaries (Hastie et al., 2009) while leading overall in Top-1 Accuracy (76.29%), Macro-F1 (76.18%), and MCC (0.5996).

Audit Evidence: Hastie et al. (2009); Slide 14.

[TIER 3: ADVERSARIAL / EDGE-CASE] Doesn't logistic regression also compute predictions via a simple vector dot product at test time?

DEFENSE: Yes, both models have fast scoring. However, LDA offers closed-form parameter estimation during batch retraining without convergence tuning, making it simpler to maintain across distributed retail edge nodes.

Audit Evidence: Hastie et al. (2009); Slide 14.

SLIDE 18 FOCUS **Slide 18: PCA Evaluation & Rejection Decision** — *Continuous PCA vs Classification Experiment & Percentage Points*

[TIER 1: FOUNDATIONAL] How much variance is explained by the first two principal components of continuous features?

DEFENSE: In continuous PCA across the 5 financial variables, PC1 explains 45.30% of variance and PC2 explains 20.22%, together capturing 65.52% of total continuous variance.

Audit Evidence: Slide 17; pca_importance.csv.

[TIER 2: INTERMEDIATE] What were the empirical results of your pre-vs-post PCA classification tournament in script 07_pca.R?

DEFENSE: Training our Champion LDA model on original features achieves 76.29% Top-1 Accuracy and 76.18% Macro-F1. When features are projected into principal components, performance drops: with 10 components, accuracy falls by 8.54 percentage points to 67.76%; with 5 components, it drops by 9.18 percentage points to 67.11%.

Audit Evidence: Slide 18; pca_pre_post_comparison.csv; 07_pca.R.

[TIER 3: ADVERSARIAL / EDGE-CASE] Why does PCA reduce classification accuracy in this application?

DEFENSE: As Jolliffe & Cadima (2016) establish, PCA constructs axes of maximal total unsupervised variance without consideration of class labels. In classification tasks, directions of maximum variance do not necessarily maximize class separability, discarding subtle boundaries between food and beverage items.

Audit Evidence: Jolliffe & Cadima (2016); Slide 18.

SLIDE 21 FOCUS

Slide 21: Time Series Demand Forecasting (Seasonal ARIMA) — *SARIMA(0,0,0)(0,1,1)[7] with drift & 30-Day Forecast*

[TIER 1: FOUNDATIONAL] What seasonal order was identified for your optimal ARIMA model?

DEFENSE: Seasonal ARIMA(0,0,0)(0,1,1)[7] with drift, operating on daily aggregated order counts with a 7-day weekly period. Weekly seasonal differencing (D=1, s=7) eliminated weekly non-stationarity, and the seasonal moving average parameter (sma1 = -0.920) captured intra-week cyclical demand.

Audit Evidence: Slide 21; timeseries_analysis_report.txt; 09_timeseries.R.

[TIER 2: INTERMEDIATE] Your dataset covers 90 days. How do you statistically justify fitting a seasonal ARIMA model on this horizon?

DEFENSE: For a weekly seasonal cycle (s = 7), 90 daily observations provide 12 full weekly cycles plus 6 days (N = 90). As Hyndman & Khandakar (2008) and Box & Jenkins (1976) establish, 10 to 15 completed cycles (70–105 observations) comfortably provide sufficient statistical power for seasonal differencing and parameter estimation.

Audit Evidence: Hyndman & Khandakar (2008); Box & Jenkins (1976); Slide 20.

[TIER 3: ADVERSARIAL / EDGE-CASE] How did you statistically verify that your ARIMA forecast is adequate?

DEFENSE: We evaluated residual autocorrelation via the Ljung-Box test at lag 14. The test yielded Q = 14.86 with p-value = 0.3982 (well above alpha = 0.05), indicating no detected residual autocorrelation at seasonal lags and confirming that the model adequately extracted the 7-day cyclical structure.

Audit Evidence: Ljung & Box (1978); Slide 22.

SLIDE 24 FOCUS

Slide 24: Industry Partner Consultation & Operational SLAs — *Mr. Kusan Edirisinghe (CEO, BladeGen Tech) & Guardrails*

[TIER 1: FOUNDATIONAL] Who is Mr. Kusan Edirisinghe and what role did he play in your consultancy?

DEFENSE: Mr. Kusan Edirisinghe is the CEO of BladeGen Tech (Pvt) Ltd, the enterprise data and POS telemetry partner for the restaurant chain. He provided expert industry validation on edge latency, checkout queue limits, and inventory integration.

Audit Evidence: Slide 24; Consultation Minutes.

[TIER 2: INTERMEDIATE] What operational guardrails were integrated into your architecture based on his feedback?

DEFENSE: Two critical operational rules: (1) An 'Express Mode' bypass to automatically suppress cashier prompt popups during peak morning rush hours (8:00–10:00 AM) to maintain queue speed (<45s per patron), and (2) a real-time stockout check to prevent suggesting items that are currently unavailable in the kitchen.

Audit Evidence: Slide 24; Slide 27.

[TIER 3: ADVERSARIAL / EDGE-CASE] Were these architectural features implemented in software or proposed as specifications?

DEFENSE: They are proposed operational specifications and business rules designed for the implementation roadmap, validated through industry consultation rather than full software deployment.

Audit Evidence: Slide 24; Slide 26.

SLIDE 27 FOCUS **Slide 27: Final Consultancy Recommendations** — *Phased Pilot, Conversion Target & PDPA Governance*

[TIER 1: FOUNDATIONAL] What are your five final recommendations for BladeGen Tech?

DEFENSE: 1. Pilot the Edge LDA recommendation model across flagship outlets to lift category conversion towards a 14.5% target. 2. Integrate Seasonal ARIMA daily demand forecasts into central dairy and bakery procurement. 3. Implement Morning Rush Express Mode to protect checkout throughput. 4. Execute a 4-week RCBD promotional trial blocking stores by foot traffic. 5. Establish comprehensive PDPA data governance with salted SHA-256 pseudonymization.

Audit Evidence: Slide 27; Executive Summary.

[TIER 2: INTERMEDIATE] How does your proposed RCBD promotional trial isolate true promotional lift?

DEFENSE: By blocking the 80 retail outlets into High, Medium, and Low baseline sales volume tiers before randomizing discount offers (Control, 10%, 20%), store-to-store baseline volume variance is partitioned from experimental error (Bradlow et al., 2023).

Audit Evidence: Bradlow et al. (2023); Slide 16.

[TIER 3: ADVERSARIAL / EDGE-CASE] Does salted SHA-256 hashing alone guarantee complete PDPA compliance?

DEFENSE: No, pseudonymization is one technical control. Full PDPA compliance requires end-to-end data minimization, documented access control policies, employee privacy training, and automated data retention schedules.

Audit Evidence: Slide 25; Slide 27.